

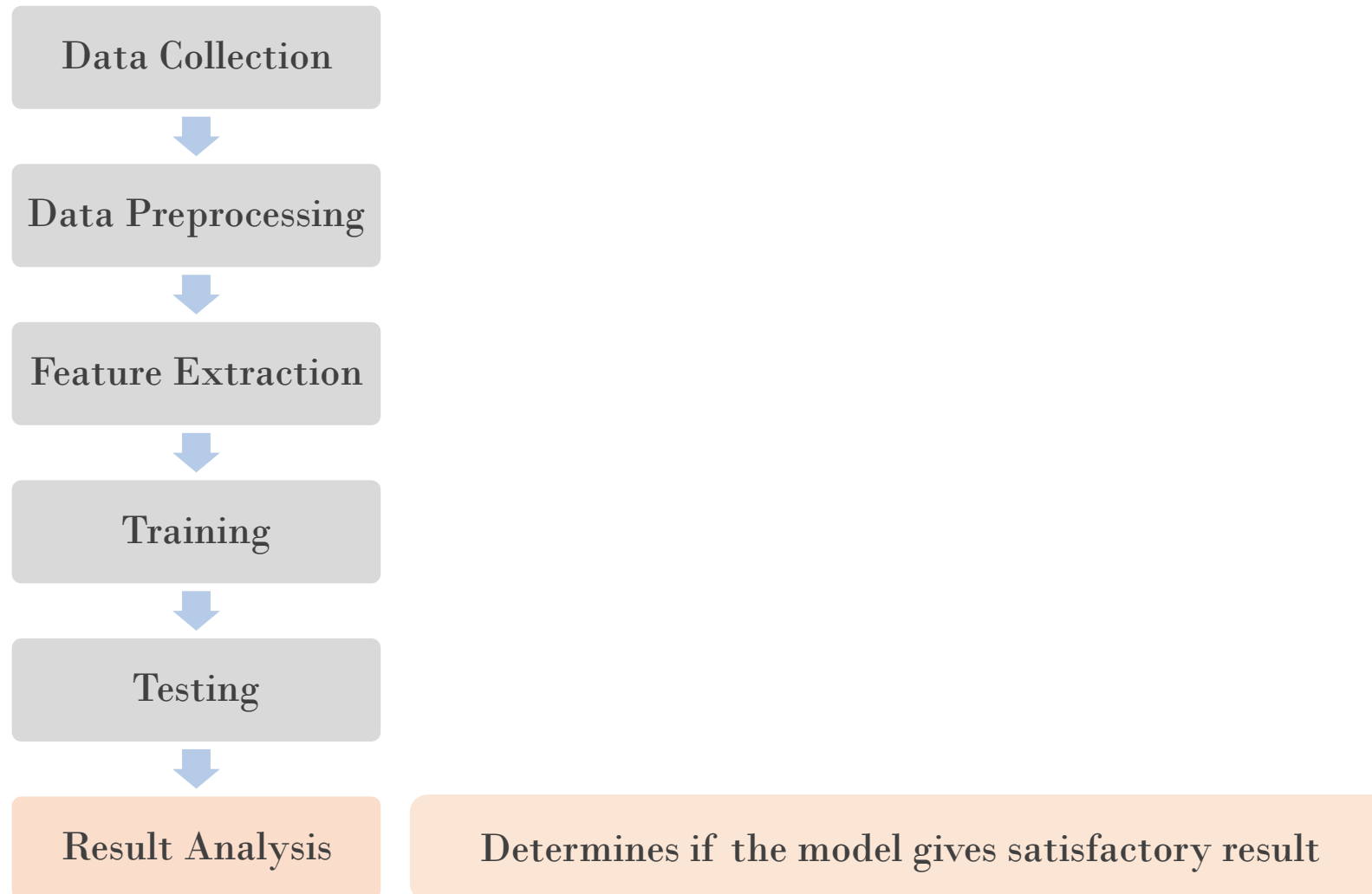
Evaluation Metrics

Data Science Practice

Outline

- False Positive Rate/Type 1 Error Rate/False Acceptance Rate
- False Negative Rate/Type 2 Error Rate/False Rejection Rate
- Detection Error Tradeoff
- Equal Error Rate
- Misclassification Cost
- Recall/Precision

Data Science Experiment Flow



Authentication Case



		Predicted Status	
		Legitimate User	Imposter
Actual Status	Legitimate User	Legitimate predicted as legitimate --- True Positive	Legitimate predicted as imposter --- False Rejection --- Type 2 Error --- False Negative
	Imposter	Imposter predicted as legitimate --- False Acceptance --- Type 1 Error --- False Positive	Imposter predicted as imposter --- True Negative

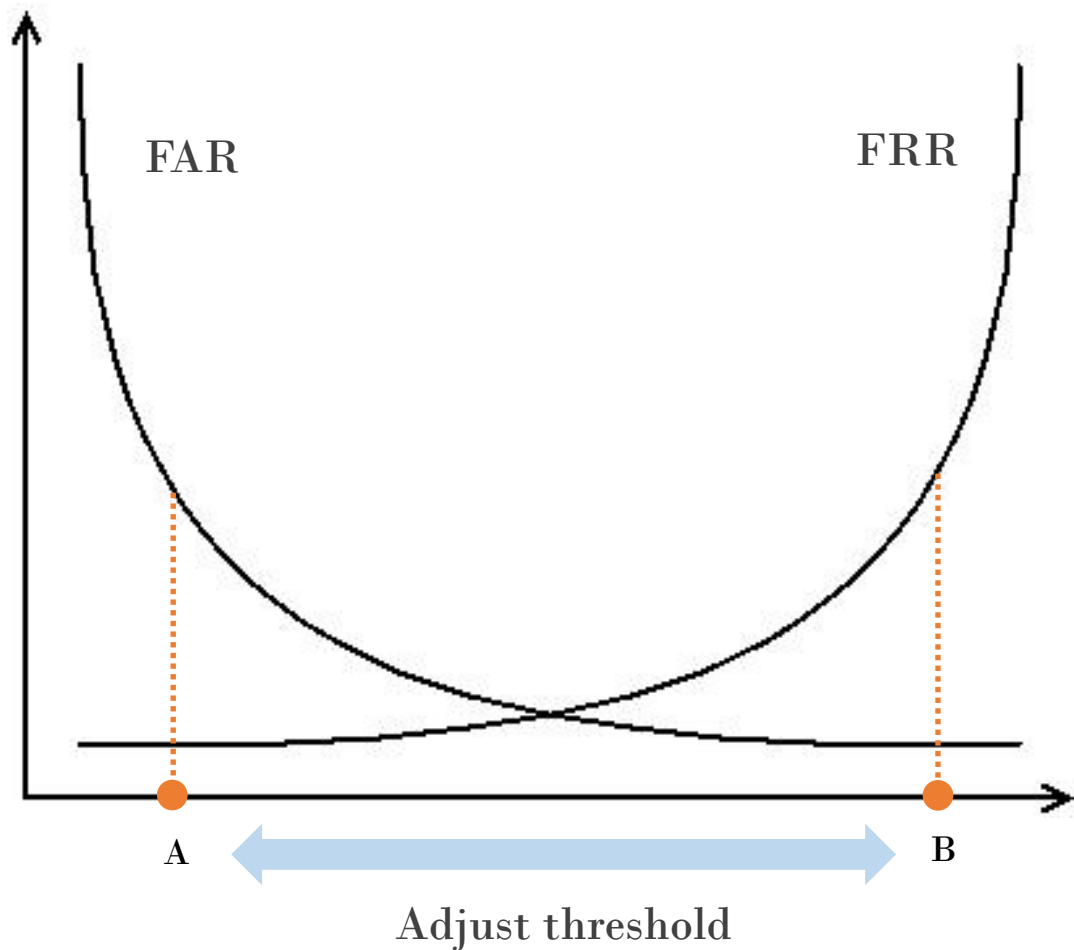
FRR & FAR

		Predicted Status	
		Legitimate User	Imposter
Actual Status	Legitimate User	TP	FN
	Imposter	FP	TN

$$\text{False Rejection Rate} = \frac{FN}{FN + TN}$$

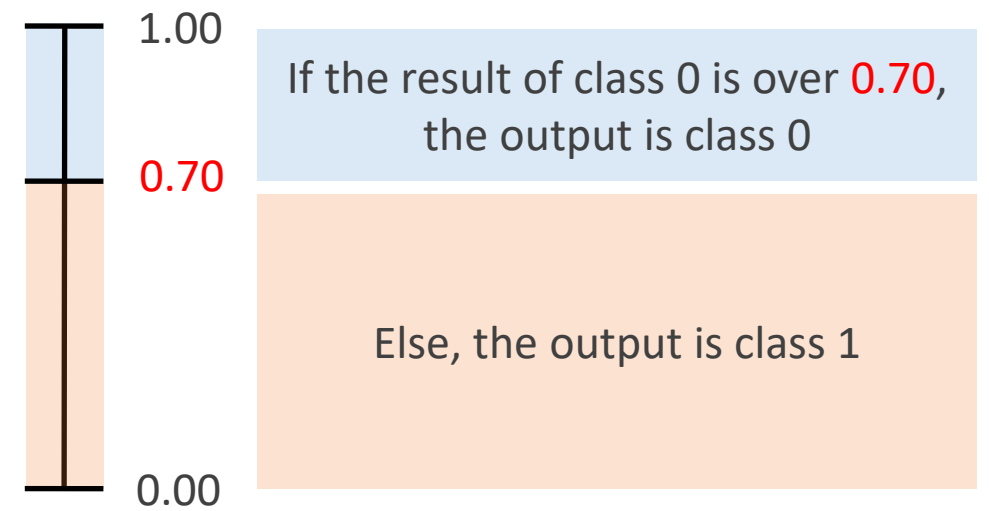
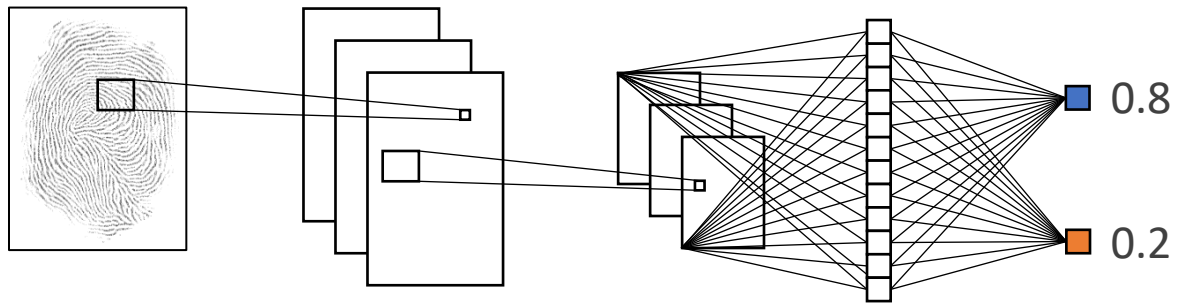
$$\text{False Acceptance Rate} = \frac{FP}{FP + TN}$$

FAR & FRR



- FAR has **negative correlation** with FRR
- Higher FAR causes lower FRR, vice versa
- High FRR = more legitimate detected as imposter, more secure system
- High FAR = more imposter detected as legitimate, more convenient for the user

Threshold



Try to find the FAR and FRR for each threshold!

Actual Class	0		0		1		1		FAR	FRR
Output	0.8 – 0.2		0.6 – 0.4		0.3 – 0.7		0.6 – 0.4			
Threshold	Predicted As	Status	Predicted As	Status	Predicted As	Status	Predicted As	Status		
0.0										
0.1										
0.2										
0.3										
0.4										
0.5										
0.6										
0.7										
0.8										
0.9										
1.0										

Try to find the FAR and FRR for each threshold!

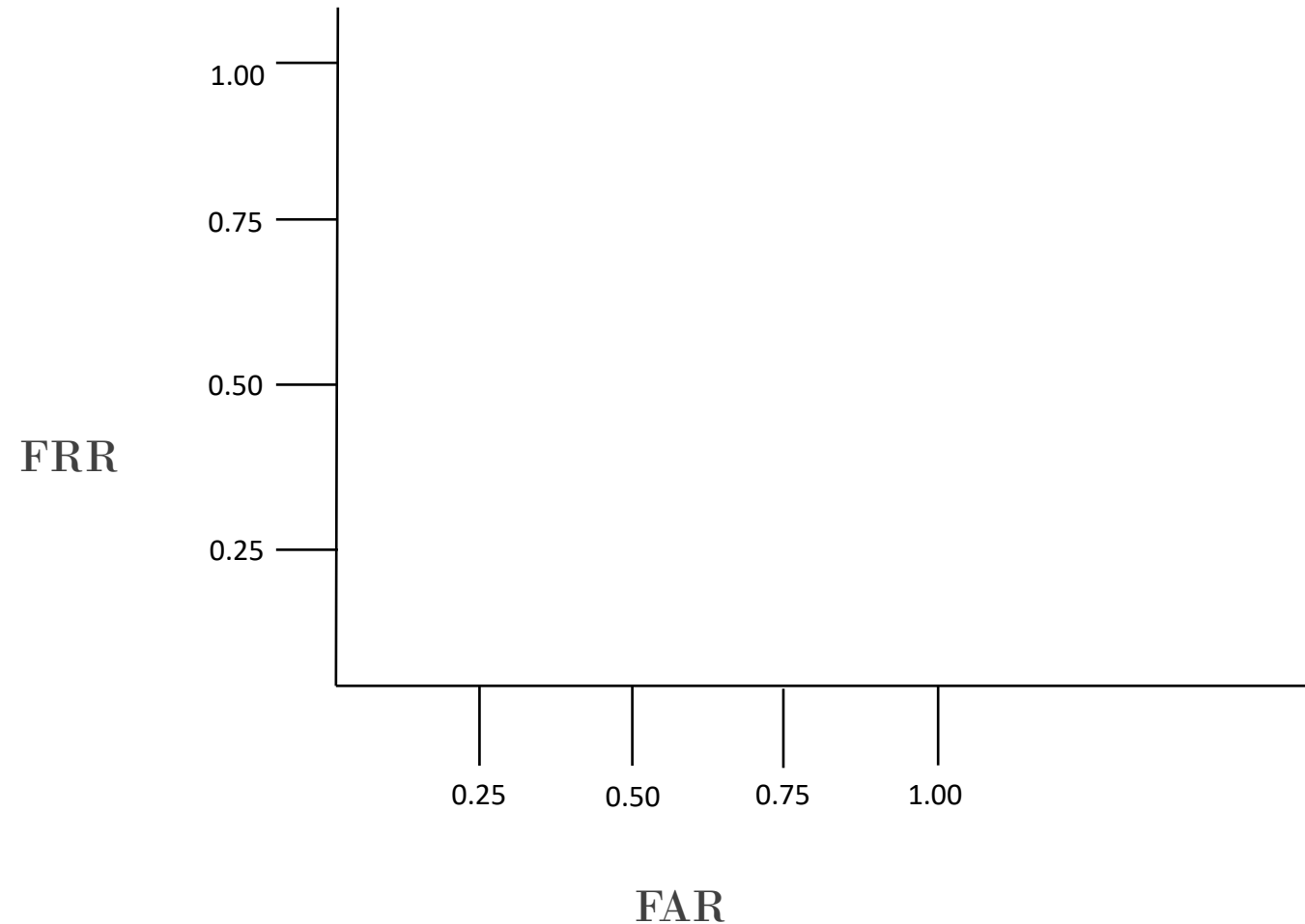
Actual Class	0		0		1		1		FAR	FRR
Output	0.8 – 0.2		0.6 – 0.4		0.3 – 0.7		0.6 – 0.4			
Threshold	Predicted As	Status	Predicted As	Status	Predicted As	Status	Predicted As	Status		
0.0	0	TP	0	TP	0	FP	0	FP		
0.1	0	TP	0	TP	0	FP	0	FP		
0.2	0	TP	0	TP	0	FP	0	FP		
0.3	0	TP	0	TP	0	FP	0	FP		
0.4	0	TP	0	TP	1	TN	0	FP		
0.5	0	TP	0	TP	1	TN	0	FP		
0.6	0	TP	0	TP	1	TN	0	FP		
0.7	0	TP	1	FN	1	TN	1	TN		
0.8	0	TP	1	FN	1	TN	1	TN		
0.9	1	FN	1	FN	1	TN	1	TN		
1.0	1	FN	1	FN	1	TN	1	TN		

Try to find the FAR and FRR for each threshold!

Actual Class	0		0		1		1		FAR	FRR
Output	0.8 – 0.2		0.6 – 0.4		0.3 – 0.7		0.6 – 0.4			
Threshold	Predicted As	Status	Predicted As	Status	Predicted As	Status	Predicted As	Status		
0.0	0	TP	0	TP	0	FP	0	FP	0.5	0
0.1	0	TP	0	TP	0	FP	0	FP	0.5	0
0.2	0	TP	0	TP	0	FP	0	FP	0.5	0
0.3	0	TP	0	TP	0	FP	0	FP	0.5	0
0.4	0	TP	0	TP	1	TN	0	FP	0.25	0
0.5	0	TP	0	TP	1	TN	0	FP	0.25	0
0.6	0	TP	0	TP	1	TN	0	FP	0.25	0
0.7	0	TP	1	FN	1	TN	1	TN	0	0.25
0.8	0	TP	1	FN	1	TN	1	TN	0	0.25
0.9	1	FN	1	FN	1	TN	1	TN	0	0.5
1.0	1	FN	1	FN	1	TN	1	TN	0	0.5

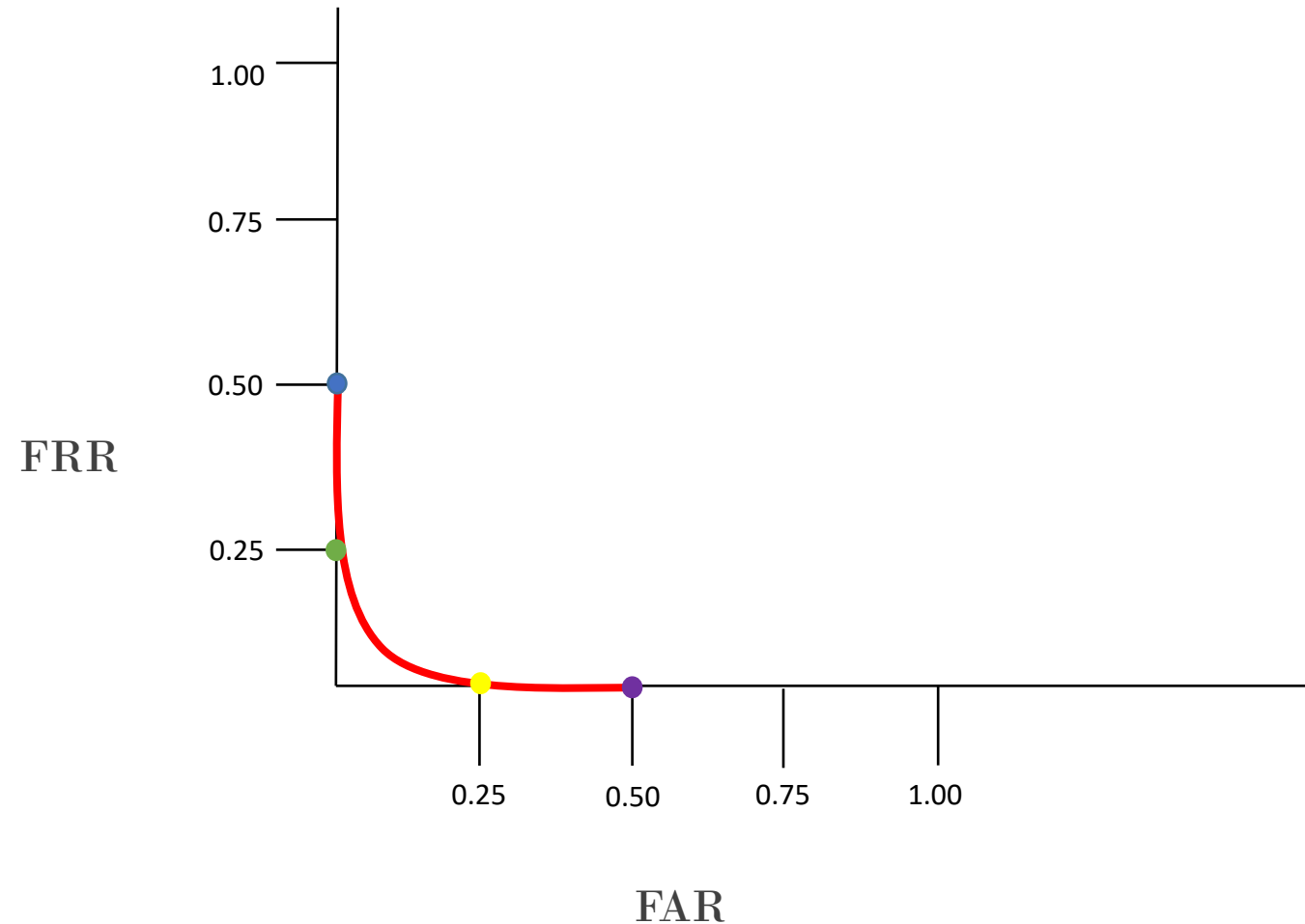
Try to Draw the DET Curve!

FAR	FRR
0.5	0
0.5	0
0.5	0
0.5	0
0.25	0
0.25	0
0.25	0
0	0.25
0	0.25
0	0.5
0	0.5

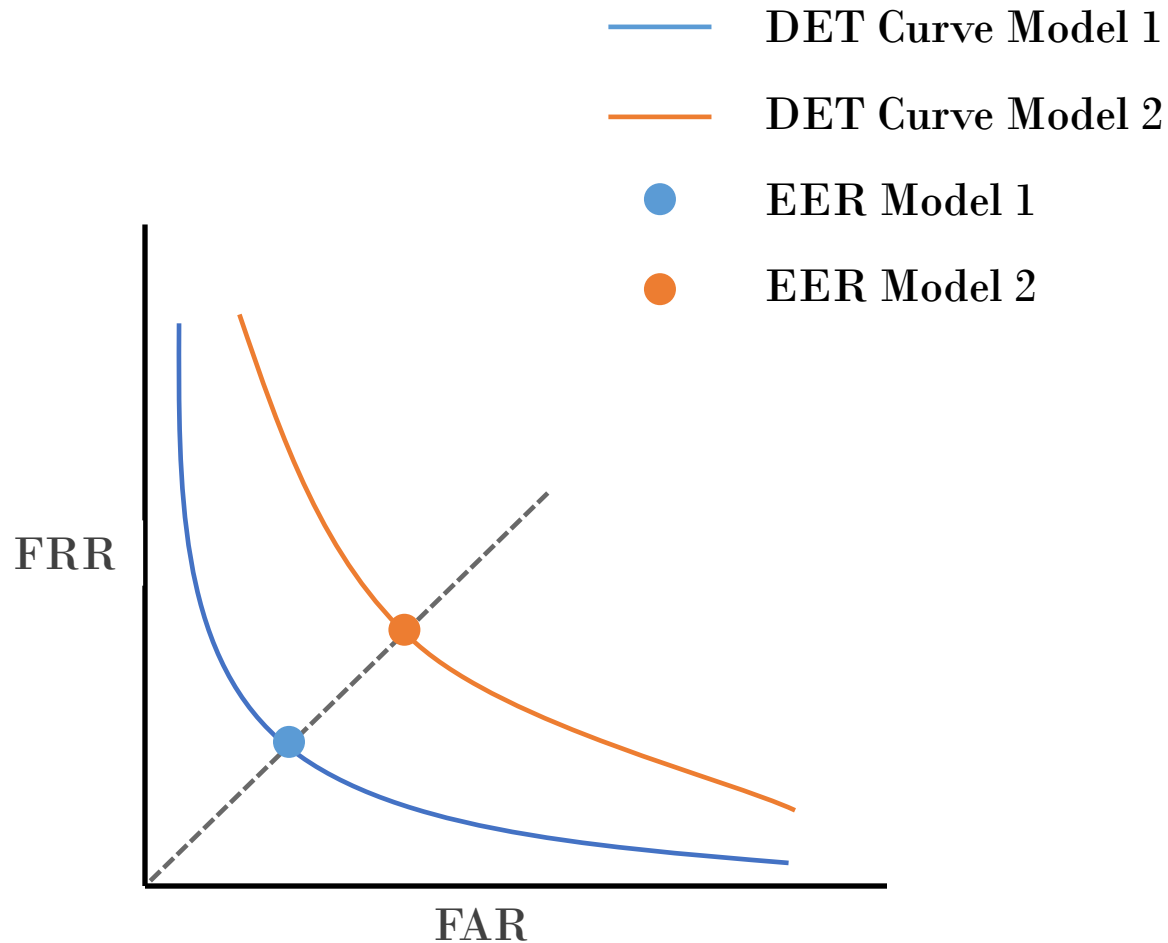


Try to Draw the DET Curve!

FAR	FRR
0.5	0
0.5	0
0.5	0
0.5	0
0.25	0
0.25	0
0.25	0
0	0.25
0	0.25
0	0.5
0	0.5

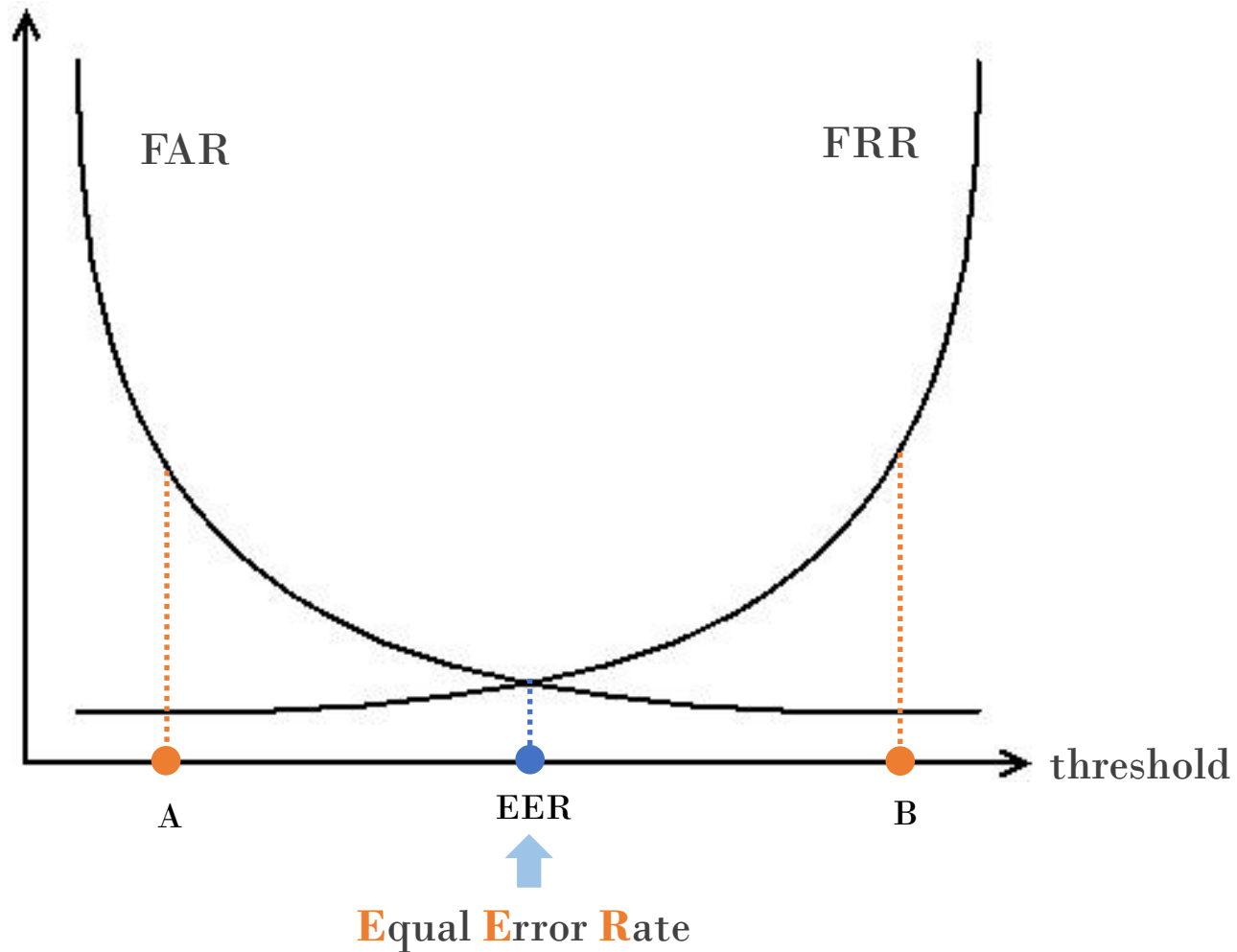


DET Curve



- **D**etection **E**rror **T**radeoff
- Maps the FAR and FRR of certain threshold
- The closer the curve to 0, the better

EER



- **EER** is a point where FAR and FRR is equal
- Minimize EER

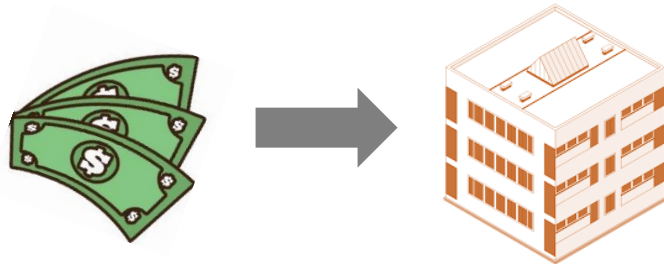
Misclassification cost

(Financial Distress Prediction Case)

False Positive/Type 1 Error/False Acceptance

Bankrupt company predicted as good

Invest \$100

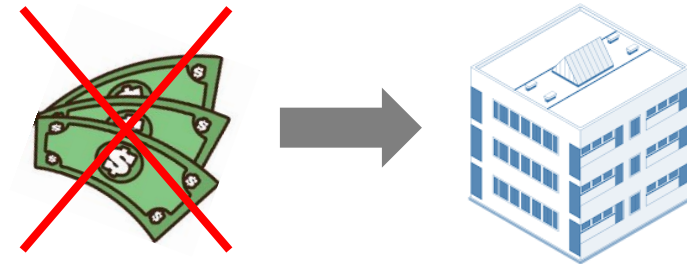


We may lose the whole **\$100**

False Negative/Type 2 Error/False Rejection

Good company predicted as bankrupt

Don't invest \$100



Suppose the dividend is 10%, **we may the lose \$10**

\$100 : \$10

Cost ratio = 10 : 1

Loss in FRR has 10 times impact than loss in FAR

Misclassification cost

(Financial Distress Prediction Case)

Prediction result on 10 companies

0 is normal company

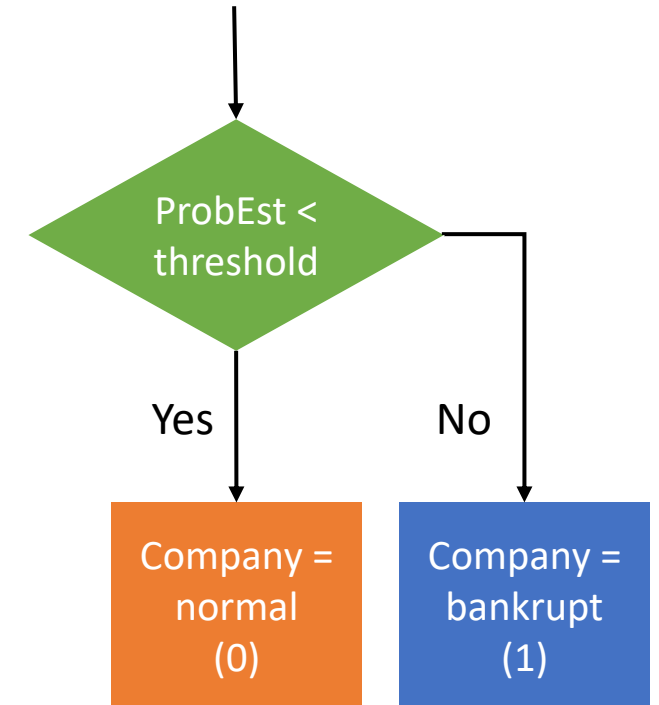
1 is bankrupt company

Company	ProbEst	Actual Class
A	0.35	0
B	0.55	0
C	0.65	0
D	0.76	0
E	0.53	1
F	0.57	1
G	0.61	1
H	0.81	1
I	0.85	1
J	0.86	1

Misclassification cost

(Financial Distress Prediction Case)

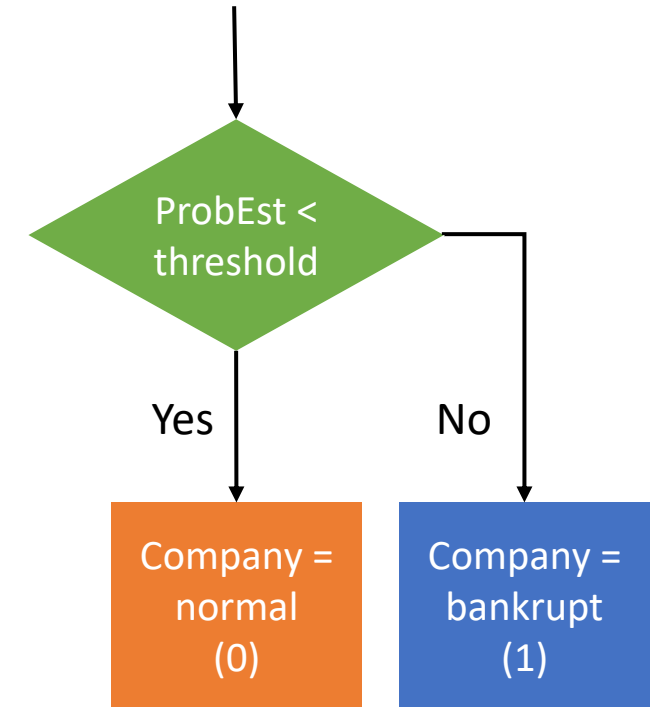
Company	ProbEst	Threshold 0.25	Threshold 0.5	Threshold 0.75	Threshold 1
A	0.35				
B	0.55				
C	0.65				
D	0.76				
E	0.53				
F	0.57				
G	0.61				
H	0.81				
I	0.85				
J	0.86				



Misclassification cost

(Financial Distress Prediction Case)

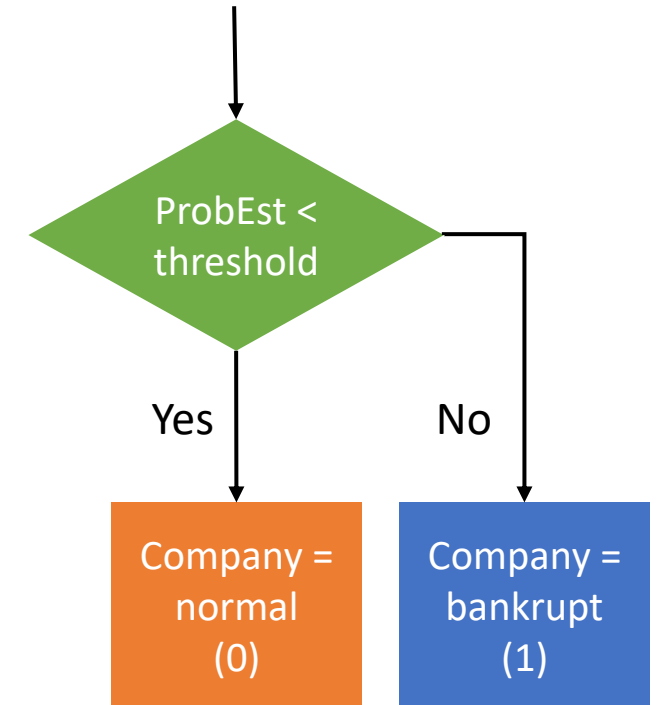
Company	ProbEst	Threshold 0.25	Threshold 0.5	Threshold 0.75	Threshold 1
A	0.35	1	0	0	0
B	0.55	1	1	0	0
C	0.65	1	1	0	0
D	0.76	1	1	1	0
E	0.53	1	1	0	0
F	0.57	1	1	0	0
G	0.61	1	1	0	0
H	0.81	1	1	1	0
I	0.85	1	1	1	0
J	0.86	1	1	1	0



Misclassification cost

(Financial Distress Prediction Case)

Company	ProbEst	Threshold 0.25	Threshold 0.5	Threshold 0.75	Threshold 1
A	0.35	1	0	0	0
B	0.55	1	1	0	0
C	0.65	1	1	0	0
D	0.76	1	1	1	0
E	0.53	1	1	0	0
F	0.57	1	1	0	0
G	0.61	1	1	0	0
H	0.81	1	1	1	0
I	0.85	1	1	1	0
J	0.86	1	1	1	0



Misclassification cost

(Financial Distress Prediction Case)

Company	ProbEst	Threshold 0.75	Actual Class	Status (TP/TN/FP/FN)
A	0.35	0	0	
B	0.55	0	0	
C	0.65	0	0	
D	0.76	1	0	
E	0.53	0	1	
F	0.57	0	1	
G	0.61	0	1	
H	0.81	1	1	
I	0.85	1	1	
J	0.86	1	1	

Compare the **prediction result**
Against the **actual class**

Positive (0) = normal

Negative (1) = bankrupt

Misclassification cost

(Financial Distress Prediction Case)

Company	ProbEst	Threshold 0.75	Actual Class	Status (TP/TN/FP/FN)
A	0.35	0	0	True Negative
B	0.55	0	0	True Negative
C	0.65	0	0	True Negative
D	0.76	1	0	False Positive
E	0.53	0	1	False Negative
F	0.57	0	1	False Negative
G	0.61	0	1	False Negative
H	0.81	1	1	True Positive
I	0.85	1	1	True Positive
J	0.86	1	1	True Positive

$$FAR = \dots?$$

$$FRR = \dots?$$

Misclassification cost

(Financial Distress Prediction Case)

Company	ProbEst	Threshold 0.75	Actual Class	Status (TP/TN/FP/FN)
A	0.35	0	0	True Negative
B	0.55	0	0	True Negative
C	0.65	0	0	True Negative
D	0.76	1	0	False Positive
E	0.53	0	1	False Negative
F	0.57	0	1	False Negative
G	0.61	0	1	False Negative
H	0.81	1	1	True Positive
I	0.85	1	1	True Positive
J	0.86	1	1	True Positive

$$FAR = \frac{FP}{FP+TN} = \frac{1}{4}$$

$$FRR = \frac{FN}{FN+TP} = \frac{3}{6}$$

Misclassification cost

(Financial Distress Prediction Case)

$$FAR = \frac{1}{4}, \quad FRR = \frac{3}{6}$$

Misclassification Cost = $(FAR * \sum Bankrupt * \text{Cost Ratio}) + (FRR * \sum Normal)$

Cost Ratio	Misclassification Cost
1	$(0.25 * 6 * 1) + (0.5 * 4) = 3.5$
2	
4	

Misclassification cost

(Financial Distress Prediction Case)

$$FAR = \frac{1}{4}, \quad FRR = \frac{3}{6}$$

$$\text{Misclassification Cost} = (FAR * \sum \text{Bankrupt} * \text{Cost Ratio}) + (FRR * \sum \text{Normal})$$

Cost Ratio	Misclassification Cost
1	$(0.25 * 6 * 1) + (0.5 * 4) = 3.5$
2	$(0.25 * 6 * 2) + (0.5 * 4) = 5$
4	$(0.25 * 6 * 4) + (0.5 * 4) = 8$

Misclassification cost

(Financial Distress Prediction Case)

Cost Ratio	Threshold	FAR	FRR	Misclassification Cost
1	0.25			
	0.5			
	0.75	0.25	0.5	3.5
	1			
2	0.25			
	0.5			
	0.75	0.25	0.5	5
	1			
4	0.25			
	0.5			
	0.75	0.25	0.5	8
	1			

Calculate misclassification cost for the **other threshold**



Company	ProbEst	Threshold 0.25	Threshold 0.5	Threshold 0.75	Threshold 1
A	0.35	1	0	0	0
B	0.55	1	1	0	0
C	0.65	1	1	0	0
D	0.76	1	1	1	0
E	0.53	1	1	0	0
F	0.57	1	1	0	0
G	0.61	1	1	0	0
H	0.81	1	1	1	0
I	0.85	1	1	1	0
J	0.86	1	1	1	0

Misclassification cost

(Financial Distress Prediction Case)

Cost Ratio	Threshold	FAR	FRR	Misclassification Cost
1	0.25	1	0	6
	0.5	0.75	0	4.5
	0.75	0.25	0.5	3.5
	1	0	1	4
2	0.25	1	0	12
	0.5	0.75	0	9
	0.75	0.25	0.5	5
	1	0	1	4
4	0.25	1	0	24
	0.5	0.75	0	18
	0.75	0.25	0.5	8
	1	0	1	4

Misclassification cost

(Financial Distress Prediction Case)

Cost Ratio	Threshold	FAR	FRR	Misclassification Cost
1	0.25	1	0	6
	0.5	0.75	0	4.5
	0.75	0.25	0.5	3.5
	1	0	1	4
2	0.25	1	0	12
	0.5	0.75	0	9
	0.75	0.25	0.5	5
	1	0	1	4
4	0.25	1	0	24
	0.5	0.75	0	18
	0.75	0.25	0.5	8
	1	0	1	4

Pick **minimum misclassification** cost as final misclassification cost

Research conducted by the Finance team
(presented later)

Compare several models in bankruptcy prediction with this baseline model

The best model is the model with minimum misclassification cost at all cost ratio.

Misclassification cost

(Financial Distress Prediction Case)

Cost Ratio	Threshold	FAR	FRR	Misclassification Cost
1	0.25	1	0	6
	0.5	0.75	0	4.5
	0.75	0.25	0.5	3.5
	1	0	1	4
2	0.25	1	0	12
	0.5	0.75	0	9
	0.75	0.25	0.5	5
	1	0	1	4
4	0.25	1	0	24
	0.5	0.75	0	18
	0.75	0.25	0.5	8
	1	0	1	4

Pick **minimum misclassification** cost as final misclassification cost

Research conducted by the Finance team
(presented later)

Compare several models in bankruptcy prediction with this baseline model

The best model is the model with minimum misclassification cost at all cost ratio.

Recall & Precision

		Predicted Status	
		Legitimate User	Imposter
Actual Status	Legitimate User	TP	FN
	Imposter	FP	TN

Recall measures the proportion of actual positive cases that the model correctly identifies out of all actual positive cases.

Precision measures the proportion of actual positive samples within those predicted as positive by the model.

$$\text{Precision} = \frac{TP}{(FN + TP)} \text{ [Predict P]}$$

$$\text{Recall} = \frac{TP}{(FP + TP)} \text{ [actual P]}$$

When the target is more concerned about the side with less category data, we select recall, precision

Recall & Precision

How to choose between recall and precision?

Example 1: Access control system [model1, model 2]
Which one is more important?

Example 1: Access control system [model3, model 4]
Which one is more important?

Model 1		Predicted Status	
		Legitimate User	Imposter
Actual Status	Legitimate User	98(TP)	2 (FN)
	Imposter	0(FP)	200(TN)

Model 2		Predicted Status	
		Legitimate User	Imposter
Actual Status	Legitimate User	100(TP)	0(FN)
	Imposter	2(FP)	198(TN)

Model 3		Predicted Status		Model 4		Predicted Status	
		Legitimate User	Imposter			Legitimate User	Imposter
Actual Status	Legitimate User	90(TP)	10(FN)	Actual Status	Legitimate User	100(TP)	0(FN)
	Imposter	0(FP)	200(TN)		Imposter	10(FP)	190(TN)

Recall & Precision

How to choose between recall and precision?

Example 1: Access control system

Which one is more important?

$$\text{Precision_model1} = 98 / 98 + 2 = 0.98$$

$$\text{Recall_model1} = 98 / 98 = 1$$

$$\text{Precision_model2} = 100 / 100 + 0 = 1$$

$$\text{Recall_model2} = 100 / 100 + 2 = 0.9802$$

Model 1		Predicted Status	
		Legitimate User	Imposter
Actual Status	Legitimate User	98(TP)	2 (FN)
	Imposter	0(FP)	200(TN)

Model 2		Predicted Status	
		Legitimate User	Imposter
Actual Status	Legitimate User	100(TP)	0(FN)
	Imposter	2(FP)	198(TN)

Recall & Precision

Example 2: Medical diagnosis
Which one is more important?

$$\text{Precision_model1} = 90 / 90 + 10 = 0.9$$

$$\text{Recall_model1} = 90 / 90 + 0 = 1$$

$$\text{Precision_model2} = 100 / 100 = 1$$

$$\text{Recall_model2} = 100 / 100 + 10 = 0.909$$

Model 3		Predicted Status	
		Legitimate User	Imposter
Actual Status	Legitimate User	90(TP)	10 (FN)
	Imposter	0(FP)	200(TN)

Model 4		Predicted Status	
		Legitimate User	Imposter
Actual Status	Legitimate User	100(TP)	0(FN)
	Imposter	10(FP)	190(TN)